3 Computational thinking: Catch the mouse



You will learn

- to use the mouse to control the computer
- to use the keyboard to control the computer
- what inputs are
- to see and hear outputs.

In this unit you will use a computer. You will use the computer mouse. You will make a cat move on the screen.



Class activity

A computer **mouse** looks like this. You can move it on the table.

Have you used a computer mouse before?

Draw a picture to show how you used the mouse.



Learning outcomes: Run and use a simple program made by someone else; Describe a program by saying what its inputs and outputs are



input output
Scratch game
mouse pointer



Talk about...

Some computers don't have a mouse. You touch the screen instead. Have you ever done that?



Did you know?

Doug Engelbart invented the computer mouse. His friends asked why he called it a mouse. He said, "Because it looks like a little mouse with a tail." Does your mouse have a tail or not?



3.1 Hello Scratch

In this lesson

You will learn:

- about **Scratch**. Scratch is a programming language for children.



This is Scratch the cat. He will help you learn.

Scratch will walk when you click with the mouse. Scratch walks inside a square. The square is called the **stage**.

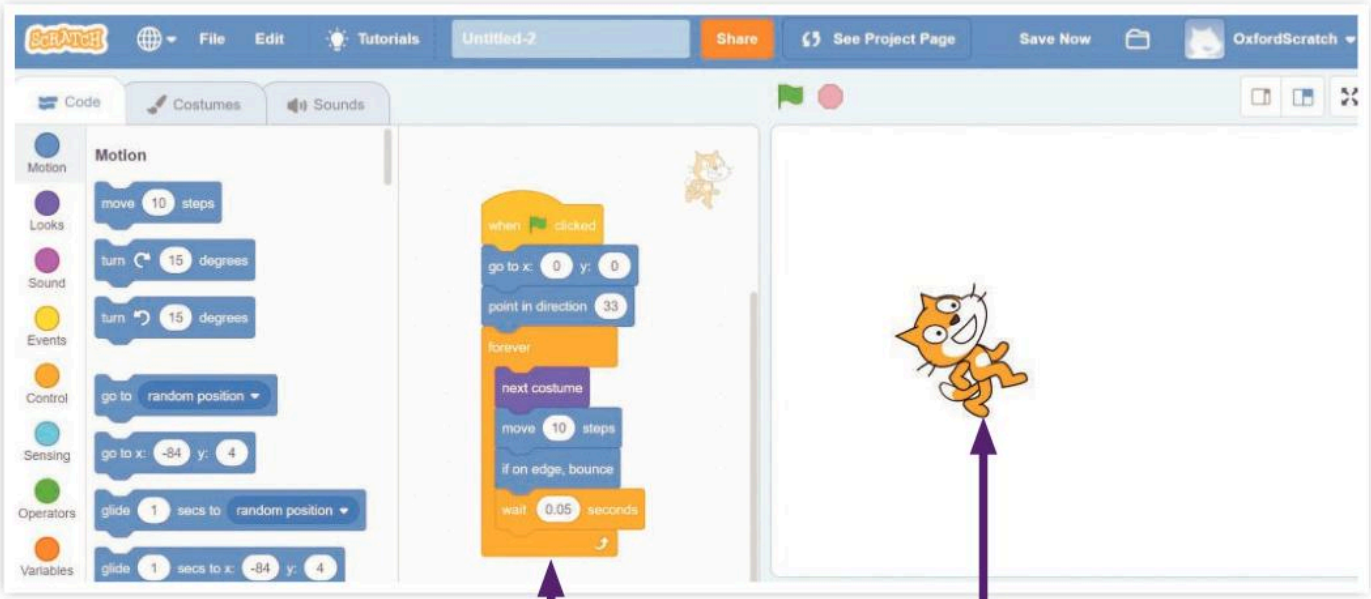


Activity

You will work with the Scratch website.

https://scratch.mit.edu/projects/editor/

A program is ready to use. The program tells Scratch to walk.



This is the program.

This is Scratch.

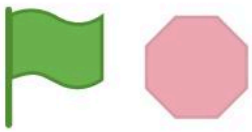
Icons are little pictures. They show you things the computer can do.

To use an icon do this:

1 Move the mouse pointer to the icon.

2 Click the icon.

There are two icons at the top of the stage. They look like this.



The green flag makes the program start. The red button makes the program stop.

1 Move the mouse pointer to the green flag. Click on the green flag to make Scratch walk.



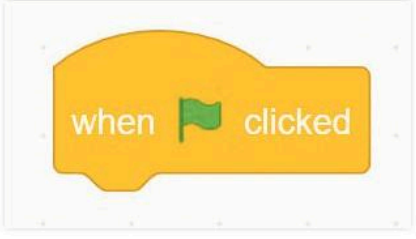
2 Move the mouse pointer to the red button. Click on the red button to make Scratch stop.



Think again

Look at the program. The program is made of blocks.

What is the picture on the first block? Why do you think the block shows that picture?



3.2 Scratch on stage

In this lesson

You will learn:

➔ how to make changes to the Scratch stage.

Last lesson you used the 'start' and 'stop' icons. By clicking the icons you could start and stop the program. Scratch walked on the stage. In this lesson you will make changes to the stage.



You can change the size of the stage.

This icon makes the stage big. Move the pointer to the icon. Click the mouse button.



This icon makes the stage small.



Click on this icon. Make the stage small again.



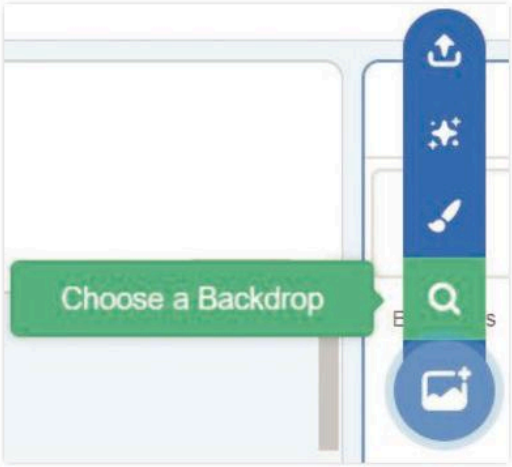
You can change the **backdrop**.

The stage can have a picture on it. The picture is called the backdrop.

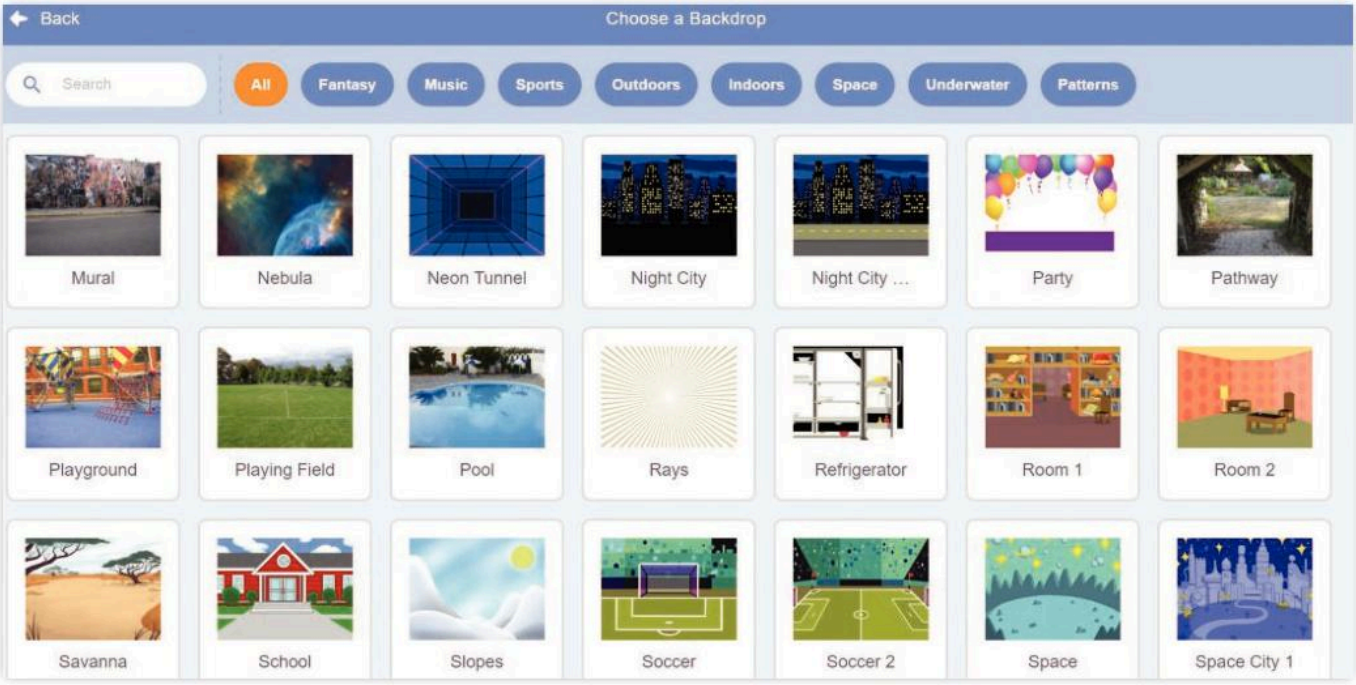


This icon lets you change the backdrop.

Click here to choose a new backdrop.



Click on any picture.



What does the stage look like now?

Be creative

Think of a new backdrop for Scratch. Paint a picture to show your idea.